



Topic: Waves

Revision Worksheet 1 MS

June 2019

1. When a guitar string is plucked, a sound of constant frequency is heard.

The wave produced on the vibrating guitar string is

- A longitudinal and progressive.
- B longitudinal and stationary.
- C transverse and progressive.
- D transverse and stationary.

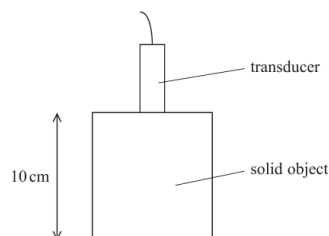
Ans :D

2. A string is stretched between two fixed points and set into oscillation. The frequency of the vibrating string is not dependent on

- A the amplitude of the string's vibration.
- B the length of the string.
- C the mass per unit length of the string.
- D the tension in the string.

Ans:A

3. An ultrasound transducer is placed on the surface of a solid object as shown



A pulse of ultrasound is transmitted into the object. The pulse is received at the transducer $50\mu\text{s}$ after it is emitted.

Which of the following can be used to calculate the speed of sound in this object?

A $\frac{0.10}{50 \times 10^{-6}}$

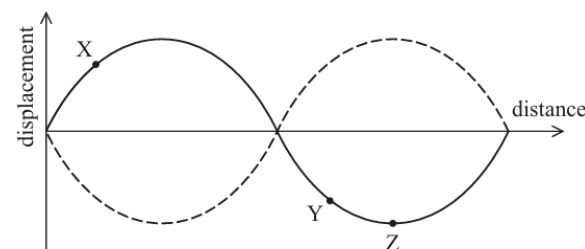
B $\frac{0.10}{25 \times 10^{-6}}$

C $\frac{0.10}{100 \times 10^{-6}}$

D $\frac{0.20}{25 \times 10^{-6}}$

Ans:B

4. The diagram shows how the displacement varies with distance along a stationary wave at two instants of time.



Which of the following statements is not correct?

- A Points X and Z are in antiphase with each other. Z
- B Points Y and Z have the same amplitude of vibration.
- C Points Y and Z have the same frequency of vibration.
- D Point Z is an antinode

Ans:B

5. Light from a discharge tube was directed through a diffraction grating, labelled as having 300 lines per mm. The diffraction pattern produced was displayed on a flat screen. The student measured the distance s between the central maximum and the first-order maximum on the screen. He also measured the distance D from the diffraction grating to the screen.

Determine whether the labelling of the diffraction grating as having 300 lines per mm was correct. (3)

$$s = 0.234 \text{ m}$$

$$D = 1.30$$

$$\lambda = 589 \text{ nm}$$

Uses $\tan\theta = s / D$ (1)
 Use of $n\lambda = d\sin\theta$ with $n = 1$ (1)
 Grating has 301 lines / mm, so the label is correct. (1)

OR

Uses $\sin\theta = \frac{s}{\sqrt{s^2 + D^2}}$ (1)
 Use of $n\lambda = d\sin\theta$ with $n = 1$ (1)
 Grating has 301 lines / mm, so the label is correct. (1)

(Use of double slit or single slit equations does not gain any credit)
 (Allow reverse calculation to show that 300 lines per mm leads to a value of λ that is close to the given value or that 300 lines per mm leads to a value of d or θ that is close to a value calculated).

Example of calculation

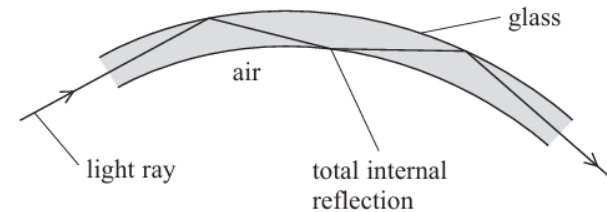
$$\tan\theta = 0.234 \text{ m} / 1.30 \text{ m} = 0.18$$

$$\theta = 10.2^\circ$$

$$n\lambda = d\sin\theta, \text{ so } d = 589 \times 10^{-9} \text{ m} / \sin 10.2^\circ = 3.33 \times 10^{-6} \text{ m}$$

$$\text{lines per mm} = 1 / 3.33 \times 10^{-3} \text{ m} = 301.$$

6. An endoscope is a medical device that can be used to see inside the digestive system. The endoscope contains bundles of very fine glass fibres. Light is directed into the fibres to illuminate the part of the digestive system being investigated. Light travels along the glass fibres by total internal reflection.



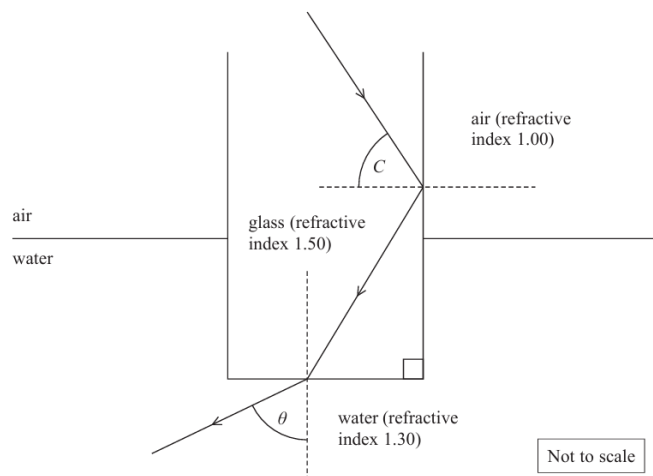
- (a) State the conditions required for total internal reflection to occur. (2)

For light travelling in a more (optically) dense substance and meeting a less (optically) dense substance Or for light travelling a material with higher RI and meeting one with a lower RI	(1)
Angle of incidence is greater than (or equal to) the <u>critical angle</u>	(1)

- (b) A ray of light is incident on the end of a glass fibre and refracts as it enters the glass. Explain why the light refracts as it enters the glass. (2)

Wave slows down as it enters the glass	(1)
Part of the wave(front) meets the glass first, so wave direction changes	(1)
OR	
Refractive index of glass is greater than that of air Or Density of glass is greater than that of air	(1)
So angle of incidence is greater than the angle of refraction (accept "bends towards the normal")	(1)

- (c) The diagram shows a ray of light being totally internally reflected at one side of a glass fibre and then refracting as it leaves the end of the fibre. The angle of incidence at the boundary between the glass fibre and air is equal to the critical angle C for the glass-air interface. The ray enters water as it leaves the glass fibre.



Determine the angle between the emerging ray and the normal. (4)

Use of $\sin C = 1/n$ for glass-air boundary	(1)
Subtracts calculated critical angle from 90°	(1)
Use of $n_1 \sin \theta_1 = n_2 \sin \theta_2$ for glass-water boundary	(1)
$\theta = 59^\circ$	(1)

(For MP3, both 1.50 and 1.30 need to be seen in the calculation).

Example of calculation

Critical angle for glass-air boundary = $\sin^{-1}(1.00 / 1.50) = 41.8^\circ$
 Angle of incidence for glass-water boundary = $90^\circ - 41.8^\circ = 48.2^\circ$
 $1.50 \times \sin(48.2^\circ) = 1.30 \times (\sin \theta)$
 $\theta = 59.3^\circ$

2019 January

7. The photograph shows waves in a ripple tank passing through a gap in a barrier.



Which of the following properties of waves is being demonstrated as the waves pass through the gap?

A diffraction

B reflection

C refraction

D superposition

8. Light travels through air and enters a glass block. Which of the following quantities does not change as the light enters the glass block?

A amplitude

B frequency

C velocity

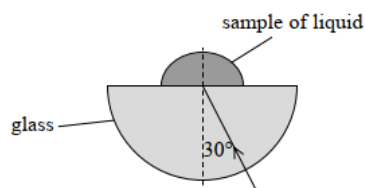
D wavelength

9. A refractometer can be used to measure refractive index. A refractometer makes use of the total internal reflection of light at a boundary between two materials.
- (a) Describe the conditions required for total internal reflection to take place at a boundary. (2)

Light must reach a boundary travelling in a medium of higher refractive index towards one of lower refractive index
Or light must reach a boundary at which velocity would increase in the second medium
Or light travelling in more dense medium towards a less dense [rarer] medium (1)
 Angle of incidence must be greater than (or equal to) the critical angle (1)

(b) One type of refractometer is shown.

A sample of liquid is placed on one side of a semicircular glass prism. A ray of light is shown passing into the prism



Determine whether this ray will be totally internally reflected from the boundary between the prism and the liquid. (3)

$$\text{liquid } \mu_{\text{glass}} = 1.15$$

Use of $\mu_1 \sin i = \mu_2 \sin r$ (1)

Either
 Critical angle = 60° (1)

The ray will not totally internally reflect as the angle of incidence is smaller than the critical angle (1)

Or
 Refracted angle = 35° (1)

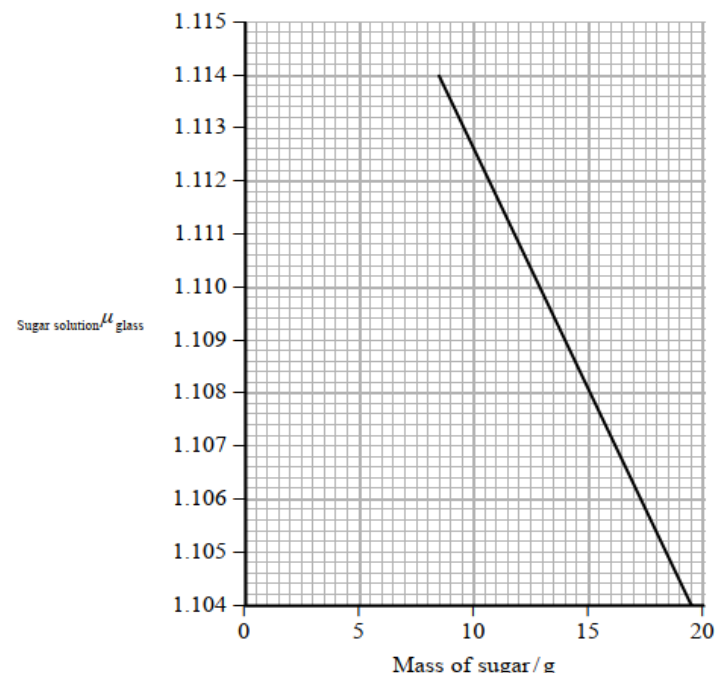
The ray does not totally internally reflect as the ray has refracted into the liquid (1)

Example of calculation

$$\frac{1}{1.15} = \frac{\sin i}{\sin 90} = 60.4^\circ$$

So not TIR as this is more than 30°

(c) The graph shows how the refractive index sugar solution μ_{glass} varies with the mass of sugar dissolved in a fixed volume of water.



Explain why the critical angle would have to be measured to a precision of more than two significant figures in order to determine the sugar content to the nearest gram. (2)

Clear link between a mass change and a change in refractive index
 e.g. A 1 gram change corresponds to ($\sim$)0.001 change in refractive index **Or** a calculation of gradient (1)

Comment that (to two sig figs) the refractive index values do not change across the range of the graph

Or
 Data used in the graph (for refractive index values) are given to more than two sig figs. (1)

10. Scientists have succeeded in levitating droplets of liquid. A standing wave is created using ultrasound generated by two sources, one above and one below the liquid. The droplets of liquid are trapped at positions corresponding to nodes as shown.



- (a) Explain how a standing wave is formed. (3)

(Two or more waves meet) travelling in opposite directions Or A wave and its reflected wave	(1)
and superpose / interfere	(1)
Creating points where the waves are in phase and points where they are in antiphase	
Or creating points of zero/min amplitude and points of maximum amplitude	(1)

- (b) Describe the motion of the air at a node and at an antinode. (3)

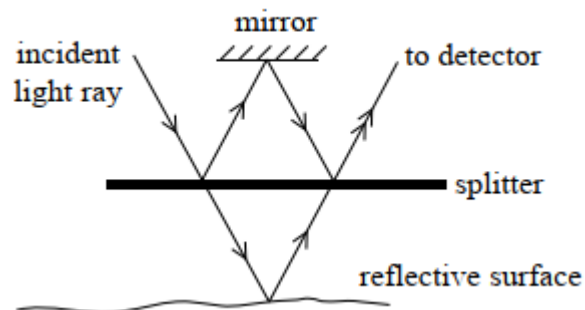
The particles (of air) are stationary Or the particles have minimum/0 amplitude/displacement at a node	(1)
The particles (of air) oscillate/vibrate with maximum amplitude at an antinode	(1)
The particles move in a direction that is parallel to the direction of the (original) wave at an antinode	(1)

- (c) The scale of the photograph is half full-size. Determine the frequency of the ultrasound wave.

speed of ultrasound in air = 320 m s^{-1} (5)

Measures distance across a minimum of 3 spaces	(1)
Scales x2	(1)
Wavelength = $2 \times$ distance between droplets	(1)
Use of $v = f\lambda$	(1)
Frequency – 40000 - 44000 Hz	(1)
<u>Example of calculation</u>	
Distance across 6 spaces = 11.5 mm	
distance between droplets = 1.9 mm	
scaled distance between droplets = $2 \times 1.9 = 3.8 \text{ mm}$	
wavelength = $2 \times 3.8 \text{ mm} = 7.6 \text{ mm}$	
$320 = f \times 7.6 \times 10^{-3}$	
$f = 41700 \text{ Hz}$	

11. In some scientific equipment it is important that surfaces are extremely flat. One method that can be used to measure the variation in the height of a reflective surface is shown in the diagram



An incident light ray is directed at a splitter. The splitter partially reflects and partially transmits the ray. The reflected ray then hits a mirror and is reflected back to the splitter. The transmitted ray hits the reflective surface and reflects back to the splitter.

The two rays then travel to a detector where superposition occurs.

Explain what is meant by superposition. (2)

When two or more waves meet [accept overlap/coincide]	(1)
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The resultant <u>displacement</u> is the sum of the individual <u>displacements</u>	(1)
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12. Ultrasound is regularly used to produce images of babies in the womb as shown.



- (a) Explain how a pulse-echo technique provides details of the position of a baby (3)

- (b) Ultrasound can also produce information about the speed of blood flow

(QWC – Work must be clear and organised in a logical manner using technical wording where appropriate)
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(Ultrasound pulse) is reflected (from surface of baby)	(1)
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Time taken t is measured	(1)
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Depth = $\frac{1}{2} vt$	(1)
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around different parts of the baby's body.

- Explain how the ultrasound can be used in this way. (4)

Doppler effect	(1)
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Ultrasound wave reflects (from moving blood)	(1)
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Frequency/wavelength of wave changes	(1)
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The greater the change in frequency/wavelength the greater the speed of the blood	(1)
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(c) Before the use of ultrasound, X-rays were used to monitor babies in the womb.

Explain why ultrasound is safer for babies than X-rays. (3)

X-rays have higher energy (than ultrasound)	(1)
Or X-rays are more penetrating	(1)
X-rays are ionising	(1)
Causing damage/mutation of cells/DNA/tissue	(1)

2018 October

13. The distance between an approaching aircraft and an airport can be determined using the pulse-echo technique with radio waves. The time between the pulse of radio waves being emitted and the reflected pulse being received is t and the velocity of radio waves in air is c . Which of the following expressions can be used to determine the distance of the aircraft from the airport?

A $2 \times c \times t$

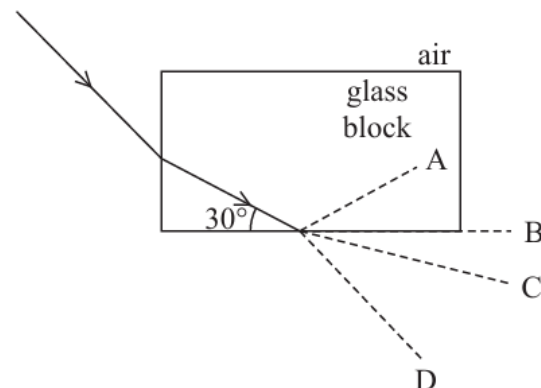
B $\frac{c \times t}{2}$

C $\frac{c}{2 \times t}$

D $c \times t$

Ans: B

14. The diagram shows the path of a ray of light into a glass block.



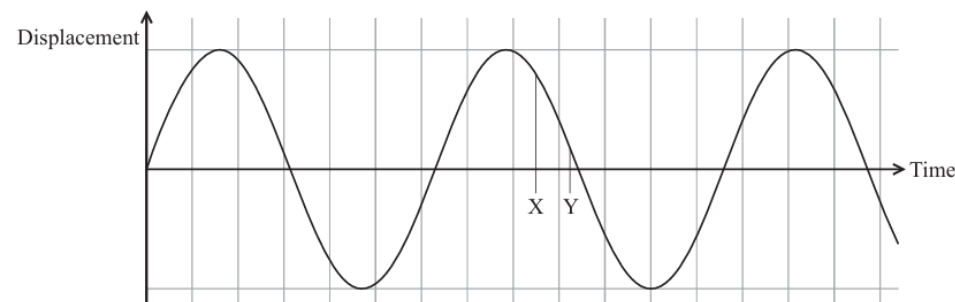
The critical angle for the glass-air boundary is 40° .

Which dotted line represents the continuing path of the ray of light?

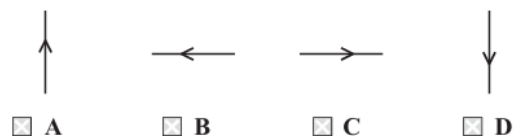
A B C D

Ans: A

15. A displacement-time graph for a particular point on a transverse wave is shown. Two times are marked X and Y.



Which of the following represents the change in displacement of this point between times X and Y?



Ans:D

16. Sound waves travel through air and pass into a concrete wall. The energy of the sound wave transmitted into the concrete wall is much less than the energy of the sound wave in air.

Which of the following wave properties is the reason for this?

- A absorption
- B diffraction
- C reflection
- D refraction

Ans:C

17. (a) Describe how a standing wave is formed. (3)

waves (travelling) in opposite directions	
Or wave and reflected wave	(1)
(The waves meet and) superpose/interfere	(1)
producing nodes and antinodes	
Or producing points of zero/minimum amplitude and points of maximum amplitude	(1)

(d) An organ pipe is open at both ends. There is an antinode at each end of the pipe.

Determine the shortest length of the pipe required to produce a sound with a wavelength of 2.6 m. (1)

1.3 m (1)

Example of calculation

$$\frac{2.6 \text{ m}}{2} = 1.3 \text{ m}$$

18. A slinky spring can be used to demonstrate a number of properties of waves. The diagram shows a wave being produced on a slinky spring when the hand oscillates from left to right.



Describe the waves that are being demonstrated in the diagram. (4)

Longitudinal wave	(1)
Compressions and rarefactions	(1)
Oscillations/vibrations parallel to the direction of wave travel	(1)
Progressive wave	
Or	
Named example of a longitudinal wave	
Or	
Description of compressions and rarefactions as areas of zero displacement	
Or	
Description of compressions as areas of high pressure and rarefactions as areas of low pressure	
Or	
Wavelength is the distance between adjacent compressions/rarefactions	(1)

19. A radio station uses a frequency of 693 kHz for its broadcasts.

(a) Explain how it is possible to receive these broadcasts in a house on the other side of a hill from the radio station.

Your answer should include an appropriate calculation. (4)

Mention of diffraction (of radiowaves) (1)

When the wavelength is of comparable size to the hill **Or** wavelength is greater than the height of the hill (1)

Use of $v = f\lambda$ with $3 \times 10^8 \text{ m s}^{-1}$ (1)

$\lambda = 433 \text{ m}$ (1)

Example of calculation

$$3 \times 10^8 \text{ ms}^{-1} = 693 \times 10^3 \text{ s}^{-1} \lambda$$

$$\lambda = 433 \text{ m}$$

(b) The radiation flux of the radio waves arriving at a particular house is $8.3 \times 10^{-9} \text{ W m}^{-2}$.

Calculate the energy arriving at a person's head from these radio waves in one hour.

area of head = 0.025 m^2 (3)

Use of Radiation flux = power per unit area (1)

Use of Energy = Power $\times$ time (1)

Energy = $7.5 \times 10^{-7} \text{ J}$ (1)

Example of calculation

$$E = 8.3 \times 10^{-9} \text{ Wm}^{-2} \times 0.025 \text{ m}^2 \times 3600 \text{ s}$$

$$E = 7.5 \times 10^{-7} \text{ J}$$

*~~(c)~~ A student makes the following comment about the answer to part (b).

"Although this electromagnetic wave energy is a very small value and safe, the same amount of electromagnetic energy at an ultraviolet frequency could be dangerous." Explain this comment. (4)

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20. A variety of colours is often seen when white light is reflected from a thin layer of oil on a puddle of water. Light reflecting from the top of this layer of oil undergoes a phase change of 180° . This results in a phase difference of 180° between the reflected light and the incident light at the surface.

(a) Explain what is meant by a phase difference of 180° . (2)

Waves are in antiphase

Or

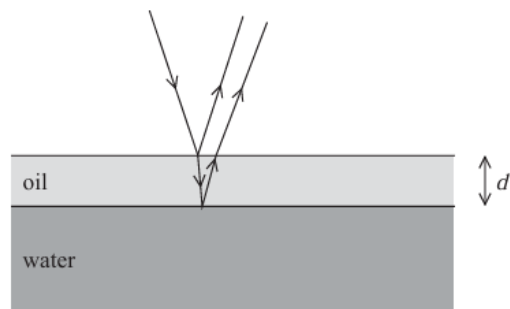
Waves are out-of-phase by half a cycle

Or

Waves have a path difference of $\frac{\lambda}{2}$ (1)

Waves would destructively interfere (1)

- (b) The diagram shows a thin layer of oil, of thickness d , on the surface of water. A ray of white light is incident on the layer of oil and reflects from the top and bottom surfaces of the oil. Light reflected from the bottom surface does not undergo a phase change



- (i) Calculate the velocity of light in oil.
refractive index of oil = 1.5 (2)

$$\text{Use of } \mu = \frac{v_1}{v_2} \text{ with } c = 3.0 \times 10^8 \text{ (m s}^{-1}\text{)} \quad (1)$$

$$\text{velocity} = 2.0 \times 10^8 \text{ m s}^{-1} \quad (1)$$

Example of calculation

$$1.5 = \frac{3 \times 10^8 \text{ ms}^{-1}}{v}$$

$$v = 2.0 \times 10^8 \text{ m s}^{-1}$$

- *(ii) When viewed from directly above the puddle, the oil is seen as yellow.
Explain this observation.

$$\text{wavelength of yellow light in oil} = 4.0 \times 10^{-7} \text{ m}$$

$$d = 1.0 \times 10^{-7} \text{ m (5)}$$

(QWC – Work must be clear and organised in a logical manner using technical wording where appropriate)

$$\text{Path difference} = 2.0 \times 10^{-7} \text{ (m) or path difference} = 2d \quad (1)$$

$$\text{This is equivalent to half a wavelength (of yellow light in oil) or } 2d = \frac{\lambda}{2} \quad (1)$$

A path difference of half a wavelength is equivalent to a phase change of 180° (or vice versa)

Or

Both waves have undergone a phase change of 180°

Or

The waves have undergone the same phase change (1)

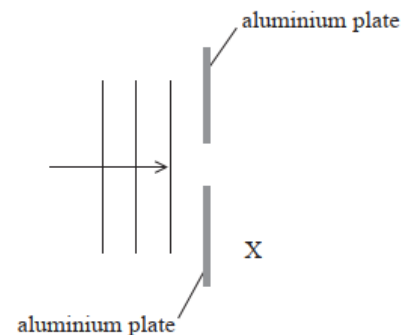
So the two waves are in phase (at the top of the oil) (1)

(and) interfere/superpose constructively (1)

2018 January

21. A teacher carried out a demonstration using microwave of wavelength 3.00 cm

- (a) The microwaves were directed towards a gap of about 3 cm between two aluminium plates as shown. Microwaves were detected at point X.



Explain why microwaves were detected at point X.

(2)

Diffraction (after passing through the gap)

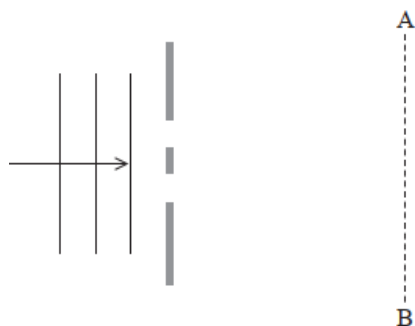
Or Microwaves spread out (after passing through the gap)

(1)

Because wavelength (of microwaves is approximately) the same as the size of the gap

(1)

(b) An extra aluminium plate was added, so the microwaves were now incident on two gaps as shown.



A microwave detector was moved along the line AB. Variations in intensity were detected. Explain why several points of maximum intensity and minimum intensity were detected along the line AB.

(5)

The idea of two (diffracted) waves

Or the idea that there are waves from both gaps

(1)

that interfere/superpose

(1)

Constructive (interference) if in phase

Or Constructive (interference) if path difference = $n\lambda$

(1)

Destructive (interference) if in antiphase

Or destructive (interference) if path difference = $(n + \frac{1}{2})\lambda$

(1)

maximum amplitude for constructive (interference)

And minimum/zero amplitude for destructive (interference)

(1)

22. A string is fixed at both ends and held under tension. When the string is plucked at its midpoint a standing wave is formed on the string. The photograph shows the string vibrating at its fundamental frequency.



(a) Explain how a standing wave is produced from travelling waves. (3)

Waves are reflected back

Or waves are (travelling) in opposite directions

(1)

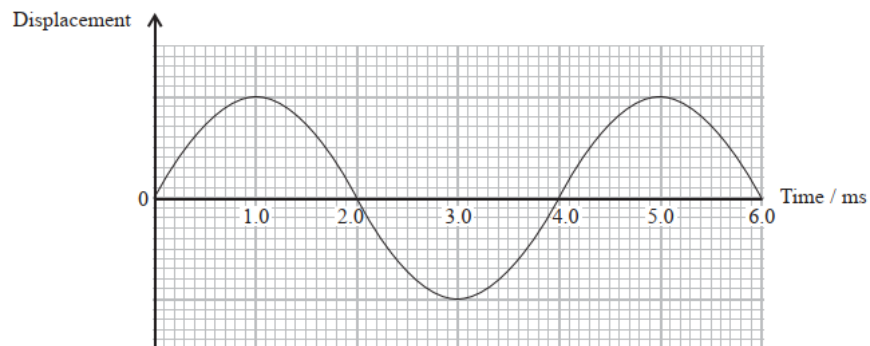
They superpose / interfere

(1)

Producing nodes and antinodes

(1)

(b) The graph shows how the displacement of the midpoint of the string varies with time as the string vibrates at its fundamental frequency.



Calculate the length of the string.

(5)

Sped of wave on string = 380 ms^{-1}

Determine time period from graph

(1)

Use of $f = \frac{1}{T}$

(1)

Use of $v = f\lambda$

(1)

Use $\lambda = 2l$

(1)

$l = 0.76 \text{ m}$

(1)

(accept use of $\lambda = vt$ or $\lambda = vT$ for MP2 and MP3 combined)

Example of calculation

$T = 0.004 \text{ s}$

$f = \frac{1}{0.004} = 250 \text{ Hz}$

$l = \frac{380 \text{ m s}^{-1}}{2 \times 250 \text{ Hz}} = 0.76 \text{ m}$

23. The photograph shows a speed gun used to measure the speed of cars.



The speed gun emits pulses of infra red radiation every 0.250 second by a moving car and then detected by the speed gun, which calculates and displays the speed of the car.

(a) The speed gun was pointed at a moving car. One pulse was detected 533 ns after being emitted. The next pulse was detected 571 ns after being emitted.

Calculate the speed of the car

(4)

Use of $s = vt$ with $v = 3.00 \times 10^8 \text{ m s}^{-1}$

(1)

Calculates difference in distances or differences in times

(1)

Correct use of factor of 2

(1)

$v = 23 \text{ m s}^{-1}$

(1)

Example of calculation

$$\Delta d = \frac{3.00 \times 10^8 \text{ m s}^{-1} \times 571 \times 10^{-9} \text{ s}}{2} - \frac{3.00 \times 10^8 \text{ m s}^{-1} \times 533 \times 10^{-9} \text{ s}}{2}$$

$\Delta d = 5.7 \text{ m}$

$v = \frac{5.7 \text{ m}}{0.25 \text{ s}} = 22.8 \text{ m s}^{-1}$

- (b) Explain why there is a maximum distance at which the speed gun is effective at measuring the speed of the car . (2)

Reflection (of pulse) is not detected (1)

(Pulse) diverges/spreads/diffracts

Or Energy/intensity (of pulse) decreases over a distance

Or Energy (of pulse) absorbed in atmosphere

Or Energy loss in reflection (of pulse)

(1)

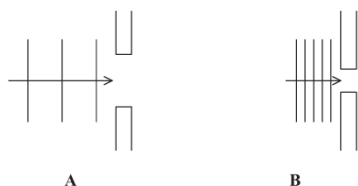
2017 October

24. The spreading of a wave as it passes around an obstacle is called

A diffraction. B polarisation. C reflection . D refraction

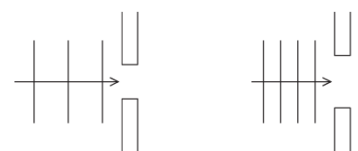
25. The four diagrams represent moving wavefronts that are one wavelength apart. In each set-up, diffraction will occur as the wavefront passes through the gap.

Which set-up will produce the greatest diffraction?



A

B



C

D

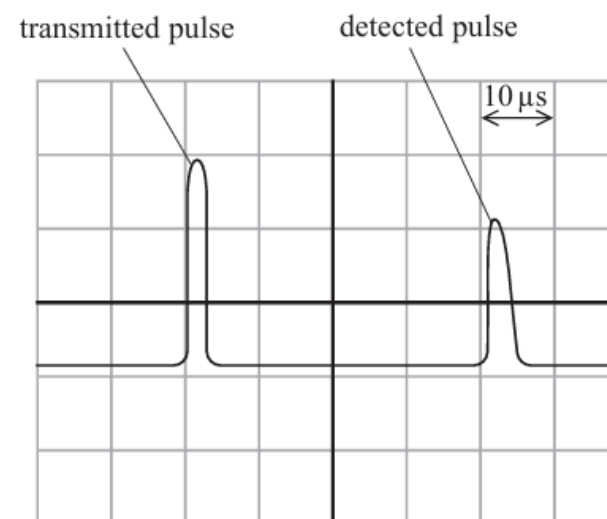
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Ans:C

Questions 26 and 27 refer to the following information

Pulse-echo techniques can use ultrasound. When the ultrasound pulse reaches the boundary between two materials part of it is reflected and then detected.

26. The oscilloscope screen below shows the transmitted and detected pulses.



The speed of ultrasound in the first material is v . Each division on the oscilloscope screen represents $10 \mu\text{s}$.

What is the approximate thickness of the first material?

A $(20 \times 10^{-6}) v$

B $(25 \times 10^{-6}) v$

C $(40 \times 10^{-6}) v$

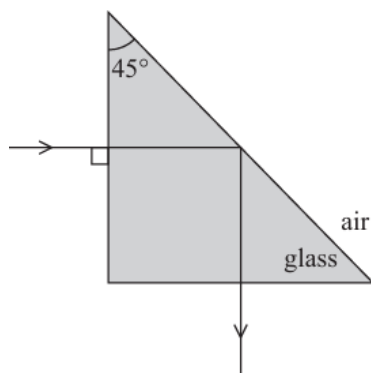
D $(80 \times 10^{-6}) v$

Ans:A

27. An ultrasound pulse-echo technique is used to produce an image by reflection from many boundaries. If the transmitted pulse is too long, the image produced is of poor quality. This could be because the
- A energy of the pulse is too high.
 - B frequency of the ultrasound is too high.
 - C time between consecutive transmitted pulses is too short.
 - D wavelength of the ultrasound is too long

Ans: C

28. A right-angled glass prism may be used in an optical device to deflect a ray of light through an angle of 90° . The light enters the prism along a normal and is totally internally reflected, as shown

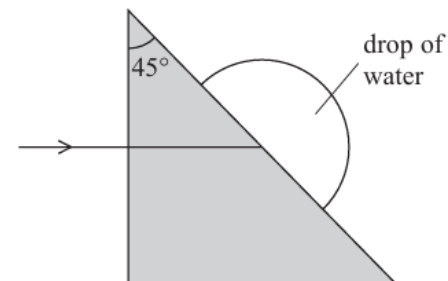


- (a) State the necessary conditions for light to be totally internally reflected at the boundary between two materials. (2)

For light travelling from a higher refractive index medium to a lower refractive index medium **Or** For light travelling from a more dense to a less dense medium (1)

Angle of incidence is greater than the critical angle (1)

- (b) A drop of water forms on the outside of the prism



Determine whether all of the light will still be deflected through 90° by the prism. (4)

speed of light in water = $2.25 \times 10^8 \text{ m s}^{-1}$

speed of light in glass = $2.00 \times 10^8 \text{ m s}^{-1}$

Use of $\frac{v_1}{v_2}$ to determine refractive index for glass-water (1)

Either

Use of ${}_1\mu_2 = \frac{\sin i}{\sin r}$ with $r = 90^\circ$ **Or** ${}_1\mu_2 = \frac{1}{\sin C}$ (1)

$C = 63^\circ$ to 64° (1)

($i < C$) so not Total Internally Reflected
(conclusion must be consistent with their calculation) (1)

Or

Use of ${}_1\mu_2 = \frac{\sin i}{\sin r}$ with $i = 45^\circ$ (1)

$r = 53^\circ$ (1)

($r < 90^\circ$) so not Total Internally Reflected
(conclusion must be consistent with their calculation) (1)

Example of calculation:

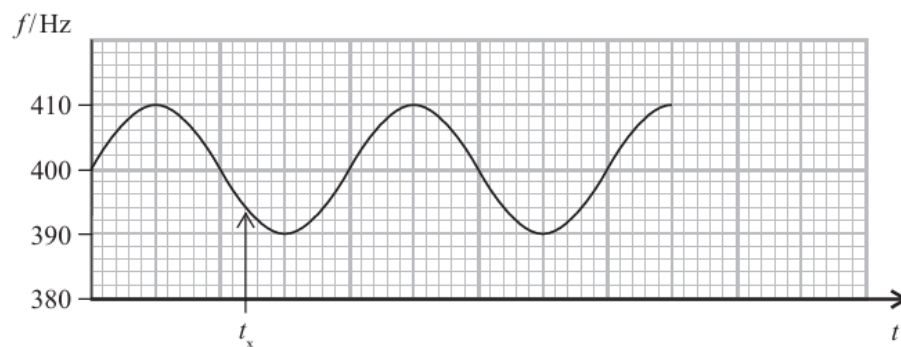
Refractive index glass to water = $\frac{2.0 \times 10^8 \text{ m s}^{-1}}{2.25 \times 10^8 \text{ m s}^{-1}} = 0.89$

$C = \sin^{-1}(0.89) = 63^\circ$

29. A teacher demonstrates the Doppler effect using a rotating source of sound. He swings a buzzer on the end of a string in a wide circle. The buzzer emits sound with a constant frequency.



A microphone connected to a computer is placed in front of the teacher to measure the observed frequency of the sound wave. A graph of observed frequency f against time t is obtained.



- (a) (i) State the frequency of sound emitted from the source. (1)

400 (Hz) (1)

- (ii) A student concludes “At time t_x the buzzer is moving away from the microphone”. Justify the student’s conclusion. (1)

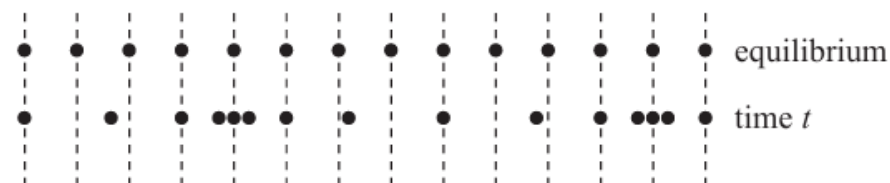
The observed frequency is less than the emitted frequency
Or observed frequency is less than 400 Hz (1)

- (b) The speed of rotation of the buzzer is increased. On the same axes, sketch the graph of f against t that would now be obtained. (2)

Greater range in frequency of audible sound centred around 400 Hz (1)

Time between high/low sounds shorter (1)

30. A longitudinal wave consists of a series of compressions and rarefactions. The diagram represents the position of the particles at equilibrium and the position of the particles at a time t .



- (a) (i) Label with a C on the diagram a position where a compression occurs at time t . (1)

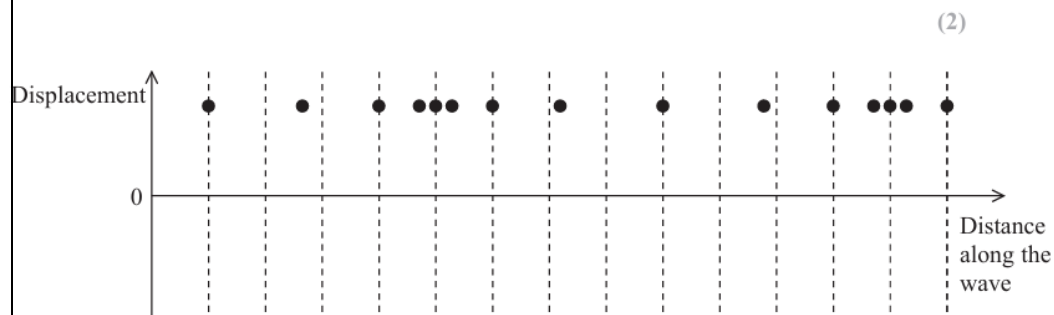
One correct position for compression labelled (1)



- (ii) Take a measurement from the diagram to determine the wavelength of the wave. (1)

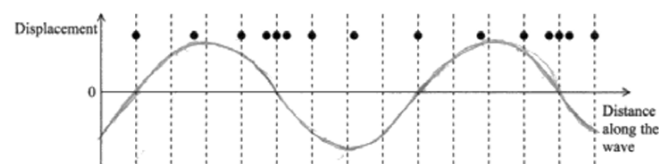
Correct wavelength 54 mm / 5.4 cm / 0.054 m (1)

- (b) A longitudinal wave can also be represented as a sine graph of the displacement of the particles against distance along the wave. On the axes below, draw a displacement-distance graph that corresponds to the longitudinal wave



Sine wave drawn with one wavelength between compressions (1)

Zero displacement at both compressions **and** the rarefaction inbetween (1)



31. (a) Microwave ovens are used to cook food. Microwaves are emitted inside the oven and reflected from the sides, forming standing waves.

- (i) One problem with using a microwave oven is the formation of hot and cold spots within the food.

Explain how hot and cold spots are formed within the food. (3)

(QWC – work must be clear and organised in a logical manner using technical terminology where appropriate)

Cold spots formed at nodes **and** hot spots formed at antinodes (1)

Either

Nodes formed where waves meet in antiphase (1)

resulting in zero/minimum amplitude (1)

Or

Antinode formed where waves meet in phase (1)

resulting in maximum amplitude (1)

(For MP2 and MP3 allow cold spot for nodes and hot spot for antinodes)

- (ii) The distance between two adjacent hot spots in the food is 1.4 cm. Calculate the frequency of the microwaves. (3)

Use of $v=f\lambda$ with $v = 3.0 \times 10^8$ (1)

Using $\lambda = 2.8$ cm (1)

$f = 1.1 \times 10^{10}$ Hz (1)

Example of Calculation:

$$f = \frac{3 \times 10^8 \text{ ms}^{-1}}{0.028 \text{ m}} = 1.07 \times 10^{10} \text{ Hz}$$

- (iii) Suggest how the effect of hot and cold spots is reduced by placing food on a rotating turntable. (1)

The positions of nodes and antinodes within the food are constantly changing (1)

Or

The standing wave is constantly changing position (1)

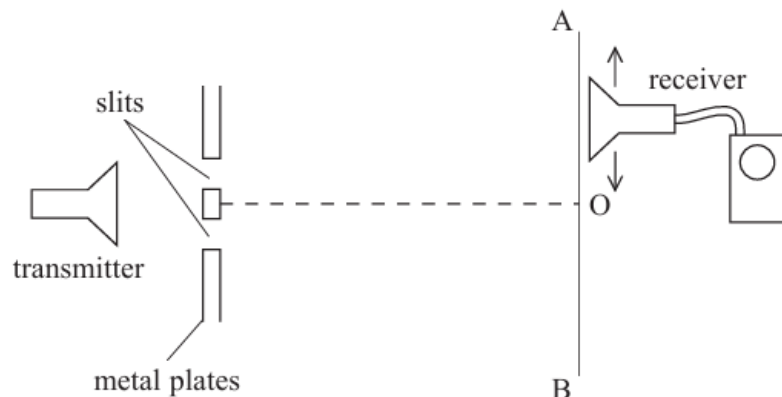
Or

Points of max **or** min amplitude are constantly changing position (1)

Or

The amplitude of the standing wave at one point in the food is constantly changing (1)

(b) The interference of microwaves can be investigated in the laboratory using a microwave transmitter and receiver. Microwaves from the transmitter pass through two slits formed between metal plates, as shown.



As the receiver is moved along AB, alternate points of maximum and minimum readings are observed. *

- (i) Explain why O is a point of maximum reading. (3)

(QWC – work must be clear and organised in a logical manner using technical terminology where appropriate)

Constructive interference (1)

Waves arrive in phase (1)

As they have zero path difference

Or point O is equidistant from both slits (1)

- (ii) Explain why maximum and minimum readings are observed alternately as the receiver is moved along the line AB. (2)

Maxima where path difference is $n\lambda$. (1)

Minima where path difference is $(n + \frac{1}{2}) \lambda$. (1)

2017 June

32. A source of sound moves at a constant speed towards a stationary observer. The frequency of the observed sound wave differs from the frequency of the emitted sound wave. The frequency of the observed sound wave is

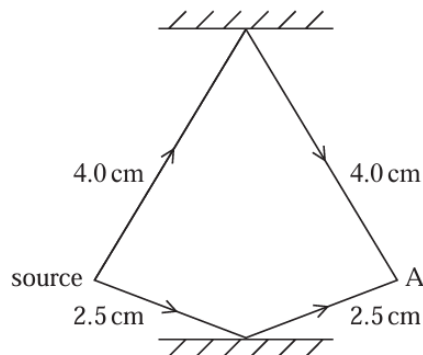
A higher because the speed of the observed wave is higher.

B higher because the wavefronts are compressed.

C lower because the speed of the observed wave is lower.

D lower because the wavefronts are spread out.

33. Two waves, each with a wavelength of 2 cm, leave a single source in phase and follow the paths shown



Which of the following is the phase difference, in radians, between the two waves as they meet at point A?

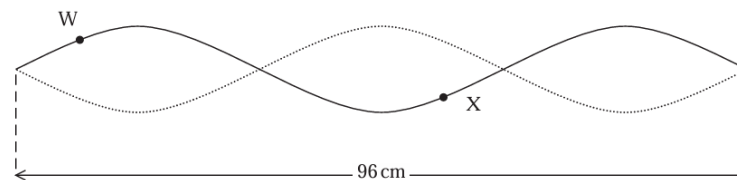
- A $\frac{\pi}{4}$
 B $\frac{\pi}{2}$
 C π
 D $\frac{3\pi}{2}$

C π radians is the phase difference between the two waves at A – the path difference between the two waves is $(8\text{cm} - 5\text{cm}) = 3\text{cm} = 1.5\lambda$. This results in the waves arriving in antiphase (π radians out of phase) which also occurs when the path difference is 1cm .

34. Which of the following quantities may be defined as the magnitude of the greatest distance of a point on a wave from the equilibrium position?

A amplitude B displacement C phase D wavelength

35. A student draws the pattern formed by a standing wave on a string.



- (a) Determine the wavelength of the wave. (1)

$\lambda = 64 \text{ cm} / 0.64\text{m} / 640\text{mm}$ (standard form alternatives accepted) (1)

- (b) W and X are points on the string. State the phase difference between W and X. (1)

Antiphase / 180° / π (radians) (1)

- (c) Describe how a standing wave is formed. (3)

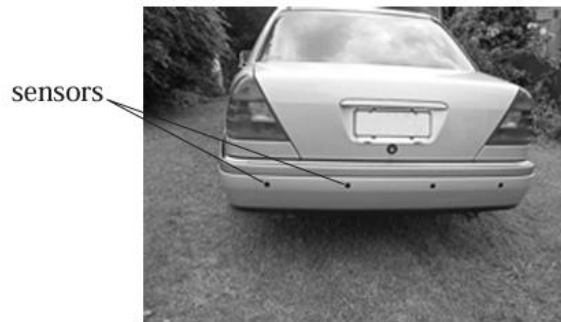
Transmitted wave and reflected wave (meet)
Or two waves travelling in opposite directions (meet) (1)

Superpose/Interfere (1)

Constructive (superposition) creates antinodes
Or Destructive (superposition) creates nodes (1)

(For constructive, accept in phase. For destructive, accept in antiphase)

36. Sensors can be fitted to the rear bumper of a car to help the driver reverse the car safely.



The sensors emit ultrasound pulses. Pulses that hit an object behind the car are reflected back to the sensors. The time taken for the reflected pulse to return is measured so that the distance to the object can be determined.

(a) State why it is necessary to emit the ultrasound in pulses. (1)

Enable pulse to return before next is emitted Or To distinguish which reflected pulse corresponds to which emitted pulse	(1)
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(b) A car manufacturer claims that the sensors are able to detect objects from a distance of 0.10 m from the car.

Calculate the maximum duration of each pulse.

speed of sound in air = 340 m s^{-1} (3)

Use of $v = \frac{s}{t}$	(1)
Correct factor of 2 (either s or their t multiplied by 2)	(1)
$t = 5.9 \times 10^{-4} \text{ s}$	(1)

Example of Calculation

$$t = \frac{0.10 \text{ m} \times 2}{340 \text{ m s}^{-1}} = 5.88 \times 10^{-4} \text{ s}$$

(c) (i) Suggest why the sensors may not help the driver when reversing towards an ascending ramp. (1)



Ultrasound is reflected away from car/sensor Or Ultrasound does not reflect (straight) back (to car)	(1)
--	-----

(ii) Suggest why the sensors may not help the driver when reversing towards a thin post. (1)

Little/no reflection (detected) Or Pulse does not hit the post Or Post is not wide enough to reflect (enough of the ultrasound)	(1)
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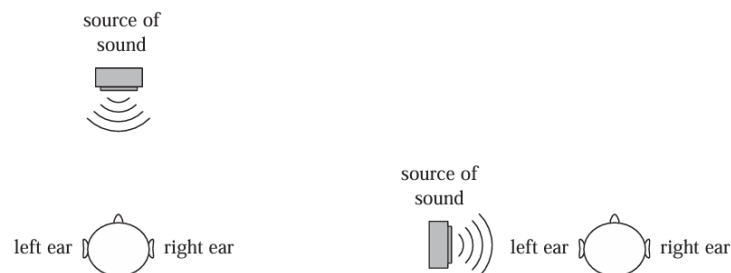
37. In a cinema, sound normally comes from a number of speakers placed at different positions around the room.

(a) Explain how sound travels through the air. (3)

Longitudinal waves	(1)
Or (Creating a series of) compressions and rarefactions	(1)
Oscillations / vibrations of (air) molecules/particles/atoms	(1)
Oscillations / vibrations parallel to the direction of the (wave) travel	(1)

(b) What a listener hears depends on the position of the source of the sound. The diagrams show a source of sound when it is placed

- in front of the listener
- to the left of the listener.



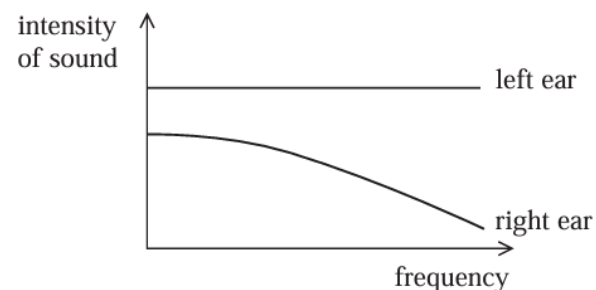
The sound waves from the source in front of the listener arrive at each ear at the same time. The sound waves from the source to the left of the listener arrive at the left ear first. The sound waves then reach the right ear by diffraction.

- (i) Describe how the sound reaches the right ear by diffraction. (2)

Spreading of a wave	(1)
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When going around head / listener / obstacle Or When passing head / listener / obstacle (MP2 dependent on MP1)	(1)
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*(ii) The intensity of a sound heard by the ear is the power per unit area reaching the ear. A particular source of sound is placed to the left of the listener. The graph shows how the intensity of the sound reaching each ear varies over a range of frequencies.



Explain the differences in the intensity of the sound reaching the left ear and the right ear. (5)

(QWC – Work must be clear and organised in a logical manner using technical wording where appropriate)

The intensity reaching the left ear is the same (for all frequencies) (1)

Right ear intensity is (always) less than the left ear intensity (1)

When sound has travelled further the energy/intensity is reduced
Or When sound has travelled further it covers a greater area
Or As sound has been reflected/absorbed by head (1)

Intensity reaching the right ear decreases as frequency increases (1)

Because higher frequencies are diffracted less
Or because shorter wavelengths are diffracted less (1)

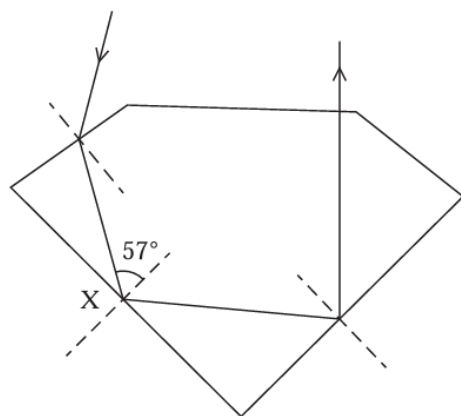
38. Diamonds are widely used in jewellery due to their ability to sparkle.

Sparkling is caused by reflections of light inside the diamond.



When used in jewellery, diamonds are cut to an optimum shape.

- (a) The diagram shows the path of one ray of light as it passes through a cut diamond.



Not to scale

Explain the path taken by the ray of light when it is incident at the diamond-air interface at X. Your answer should include a calculation.

refractive index of diamond = 2.4 (4)

Use of $n_1 \mu_2 = \frac{\sin i}{\sin r}$ with $i = 90^\circ$ **or** $\sin c = \frac{1}{\mu}$ (1)

Critical angle = 25° (accept $c = 25^\circ$) (no unit error) (1)

Recognises that angle of incidence > critical angle (allow $i > c$) (1)

Total Internal Reflection takes place (allow TIR) (1)

Example of calculation

$$c = \sin^{-1} \frac{1}{2.4} = 24.6^\circ$$

- (b) Zircon is a popular, cheaper alternative to diamond.

refractive index of diamond = 2.4 refractive index of zircon = 2.0

- (i) Explain how a higher refractive index causes diamond to sparkle more than zircon. (3)

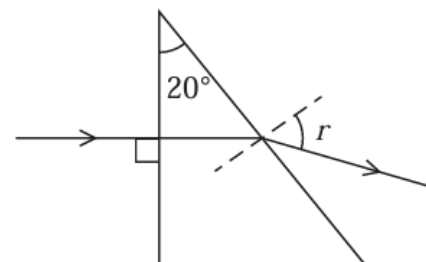
Higher refractive index corresponds to a smaller critical angle
Or see calculated value of 30° for critical angle of zircon (1)

Greater range of angles of incidence for which reflection takes place in diamond (1)

More reflection(s) for the diamond
Or more likely for reflections in the diamond
Or less refraction for the diamond (1)

(MP3 is dependent on awarding either MP1 or MP2)
(Allow converse arguments in terms of zircon for MP2 and MP3)

- (ii) The refractive index of a transparent material can be measured using a 20° prism of the material.



Not to scale

A ray of light is incident perpendicularly to one face and passes straight into the prism. The angle of refraction r , as it leaves the prism, can be measured and the refractive index of the material calculated.

By making suitable calculations suggest whether it is possible to distinguish between diamond and zircon using a protractor to measure the angles. (3)

Use of $\mu = \frac{\sin i}{\sin r}$ with 20° as one of the angles. (1)

$r(\text{diamond}) = 55^\circ$ and $r(\text{zircon}) = 43^\circ$

Or Calculates difference between $r(\text{diamond})$ and $r(\text{zircon}) = 12^\circ$ (1)

Protractor measures to $\pm 1.0^\circ$ so can be easily distinguished (MP3 is dependent on MP2 being awarded) (1)

Example of calculation

$$\sin^{-1}(2.4 \times \sin 20^\circ) - \sin^{-1}(2.0 \times \sin 20^\circ) = 12^\circ$$

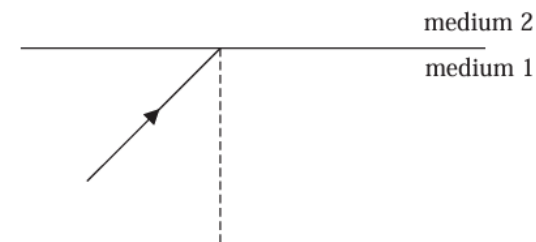
2017 January

39. Which of the following would improve the level of detail in an ultrasound scan?

- A increasing the duration of the pulses
- B increasing the frequency of the ultrasound
- C increasing the interval between the pulses
- D increasing the wavelength of the ultrasound

Ans:B

40. The diagram shows a ray of light in medium 1 striking a boundary with medium 2. In medium 1, the light has a frequency f_1 and wavelength λ_1 . In medium 2, the light has a frequency f_2 and wavelength λ_2

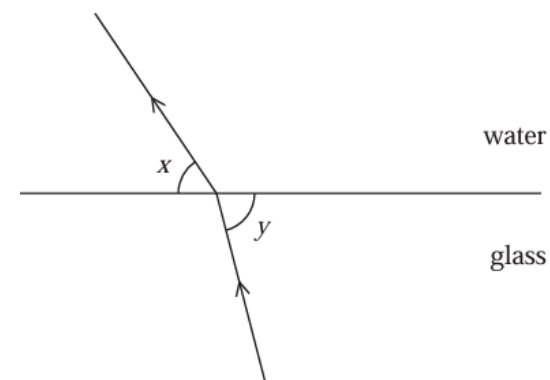


Which of the following is necessary for total internal reflection to occur?

- A $f_2 > f_1$
- B $f_1 > f_2$
- C $\lambda_2 > \lambda_1$
- D $\lambda_1 > \lambda_2$

Ans:C

41. The diagram shows a ray of light passing from glass to water.



Which of the following is a correct expression for the refractive index for light passing from glass to water?

A $\frac{\sin x}{\sin y}$

B $\frac{\sin y}{\sin x}$

C $\frac{\sin(90^\circ - x)}{\sin(90^\circ - y)}$

D $\frac{\sin(90^\circ - y)}{\sin(90^\circ - x)}$

Ans:D

42. A water wave meets a small gap in a barrier. Which row of the table correctly describes the speed and wavelength of the water wave after passing through the gap?

	speed	wavelength
☐ A	unchanged	decreased
☐ B	decreased	unchanged
☐ C	increased	increased
☐ D	unchanged	unchanged

Ans:D

43. The wavelength of a progressive wave is defined as the distance between adjacent points on the wave which
- A have the same displacement.
 - B have the same amplitude.
 - C are in phase.
 - D are in antiphase.

Ans:C

44. The photograph shows a Blue Morpho butterfly

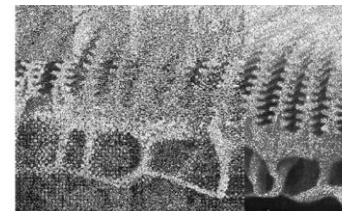


The wings appear bright blue, but they do not contain blue pigment. The blue colour is caused by superposition of the light reflected from the scales on the wings.

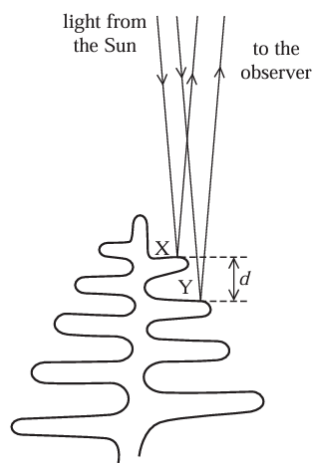
- (a) State what is meant by the principle of superposition. (2)

Idea that two (or more) waves meet	(1)
(resultant) <u>displacement</u> is sum of individual <u>displacements</u>	(1)

- (b) The scales on the wings are a few micrometres in size and have a series of ridges from which light is reflected.



White light reflects from the ridges X and Y. This causes a path difference between the light reflected from X and the light reflected from Y, as the light reaches the observer as shown.



* (i) Explain why some wavelengths of light will be observed strongly and others will not be observed. (5)

(QWC – Work must be clear and organised in a logical manner using technical wording where appropriate)

(When) path difference = $n\lambda$ (waves are) in phase
Or (when) path difference = $(n + \frac{1}{2})\lambda$ (waves are) in antiphase (1)

Constructive interference when path difference = $n\lambda$
Or constructive interference when waves in phase (1)

Destructive interference when path difference = $(n + \frac{1}{2})\lambda$
Or destructive interference when wave in antiphase (1)

Maximum amplitude when constructive interference occurs
Or zero amplitude when destructive interference occurs (1)

Constructive interference results in wavelengths that are observed (strongly)
Or destructive interference results in wavelengths that are not observed (1)

(ii) The table shows approximate wavelength ranges for colours of the spectrum.

colour	violet	blue	green	yellow	orange	red	
wavelength / nm	380	450	495	570	590	620	750

With the aid of a calculation, show that the colour observed would be blue for a butterfly where the distance d between ridges X and Y is 2.3×10^{-7} m. (2)

Evidence of use of $2 \times d$ (1)
Wavelength for max amplitude = 460 nm (blue) (1)

Example of calculation

$$\lambda = 2 \times 2.3 \times 10^{-7} \text{ m} = 4.6 \times 10^{-7} \text{ m} = 460 \text{ nm}$$